

QT 408: Introduction to Quantum Materials

Course Code: QT 408

Course Name: Introduction to Quantum Materials

L-T-P-C: 3-0-0-3

Prerequisites: Engineering Mathematics (Linear Algebra, Complex algebra, basics of 2nd of ODEs and initial value problems, 2nd order PDEs and boundary value problems, Probability and Statistics, Random variables). Maxwell's equations and EM theory at the level of the core physics syllabus from AICTE model

Intended for: UG/PG/PhD

Mutual Exclusion: PH 523, EP 301

Distribution: Elective PG/Elective UG

Semester: Odd/Even

Approval: 47th Senate

Preamble

Students of this course learn (i) The basic idea of quantum materials, (ii) The basics of band theory of solids, (iii) The basics of magnetism, (iv) The basics of superconductivity, (v) About new 2D materials like graphene, TMDCs, (vi) About topology and topological phases of matter

Course Content and syllabus

- **Band theory basics**
 - Metals, Semiconductors and Insulators
 - Band structure of solids
 - Survey of semiconducting devices for quantum technologies (electronic, quantum optical devices and principle of operation)
- **Correlated systems**
- **Magnetism**
 - Para, ferro magnetism basics
 - Magnetic measurements, hall effect, magnetoresistance
 - Faraday and Kerr effects
- **Superconductivity**
 - BCS theory
 - Ginzburg Landau
 - Josephson Effect – AC and DC Josephson effects
 - Survey of superconducting devices for quantum technologies
- **2D materials**
 - Graphene and its properties – single and few layers
 - Transition Metal Dichalcogenides – Electronic and Optical Properties

- **Topological Phases of matter**

- Basics of Topology
- Geometric phases - Berry Phase
- Aharonov Bohm effect
- Topological phases of matter

- **Survey of material growth techniques**

- Molecular beam epitaxy
- Chemical vapor deposition, MOVPE
- Pulsed laser deposition, etc.
- Crystal growth techniques

Course Outcomes

Students of this course learn

1. The basic idea of quantum materials
2. The basics of band theory of solids
3. The basics of magnetism
4. The basics of superconductivity
5. About new 2D materials like graphene, TMDCs
6. About topology and topological phases of matter

Course References

1. *Condensed Matter Physics*, M P Marder, 2nd Edition, John Wiley and Sons, 2010
2. *Introduction to Superconductivity*, Michael Tinkham, standard ed., Medtech (2017)